

Multiple Choice (10 marks)

1. Which of the following contributes to reduced energy expenditure in older adults?
 - (a) A reduction in lean body mass
 - (b) A reduction in physical activity
 - (c) A reduction in basal metabolic rate
 - (d) All of the above
2. Infants and children with a suspected food allergy e.g. cow's milk allergy/intolerance could display the following symptoms and signs:
 - (a) severe atopic dermatitis
 - (b) enterocolitis
 - (c) diarrhoea
 - (d) all of the above
3. How is iron transported in the circulation from the intestine to the sites of metabolism in the body?
 - (a) As simple Fe^{2+} in the serum
 - (b) Bound to albumin
 - (c) Bound to ferritin
 - (d) Bound to transferrin
4. Insulin deficiency is associated with
 - (a) Reduced lipolysis
 - (b) Increased ketogenesis
 - (c) Reduced gluconeogenesis
 - (d) Reduced proteolysis
5. When using bioelectrical impedance for measuring body composition, why does an electrical current travel faster through muscle than it does through fat?
 - (a) Muscle contains less water than fat
 - (b) Muscle contains more water than fat
 - (c) Muscle weighs more than fat
 - (d) Muscle weighs less than fat

True / False (10 marks)

Answer True or False

6. True or False: The sarcomere is the basic functional unit of skeletal muscle tissue.
7. True or False: Leptin acts on the hypothalamus to increase feelings of satiety, while ghrelin stimulates hunger.
8. True or False: The weighed diary is the most commonly used dietary assessment method in epidemiological research.
9. True or False: A food additive is considered safe if its Estimated Daily Intake (EDI) is less than its Acceptable Daily Intake (ADI).
10. True or False: The main metabolites of alcohol metabolism are acetaldehyde and acetate.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss Engel's law in the context of food economics and analyze how changes in income levels affect the proportion of income spent on food.
12. Discuss the differences in iodine content between organic and non-organic milk, and analyze the implications of these differences on health and nutrition.

End of Paper